Twin-Row Doubles: Adding Two Equal Groups

Doubles & Fact Strategies • Standard: CCSS.MATH.1.OA.C.6

1. MATHEMATICAL CONCEPT & STACK APPROACH: Model doubles physically using matching digit blocks in Row 1 and Row 0 of the Place Value Tray and connect to equal groups.

Doubling a quantity means adding two identical copies of the same number (n + n = 2n). Seeing matching quantities in the top and bottom rows of the tray anchors mental doubles facts.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Two-Row Tray & Digit Blocks)

Hands-On Setup: Place a [6] Digit Block in Row 1 (top) and a matching [6] Digit Block in Row 0 (bottom) of the Ones column on the Place Value Tray.

Student Prediction (Before Calculator): Observe the matching rows: 6 + 6 = 12 (1 Ten and 2 Ones on the tray).

3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [6] [ENTER] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (Symmetry & Twin Rows: Why Doubles Anchor Mental Math)

From two hands of 5 fingers (5 + 5 = 10) to two rows of 6 eggs in a carton (6 + 6 = 12), symmetry makes doubles the fastest facts in human memory. Once a student knows 7 + 7 = 14 by sight on the two-row tray, neighbor facts like 7 + 8 = 15 and 7 + 6 = 13 become instant one-step adjustments.

Core Investigation (Student Warm-Up A):

Place [7] in both rows of your Place Value Tray to build a twin-row double. What is the sum of two 7s (7 + 7)?

Near-Doubles: Doubles Plus One

Fact Fluency & Near Doubles • Standard: CCSS.MATH.2.OA.B.2

1. FOLLOW-UP EXTENSION CONCEPT, BLOCKS & PREDICTION

Concept & Stack Approach: A near-double (such as 7 + 8) is a double plus or minus 1 ((7 + 7) + 1). Evaluating the known double first (14) makes adding the remaining +1 effortless.

Blocks & Prediction: Place [7] in Row 1 (top) and [8] in Row 0 (bottom) of the Place Value Tray. Prediction: 7 and 8 is double 7 plus 1 more: 14 + 1 = 15.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[7] [ENTER] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[7]	**7**	Enter base number 7
[ENTER]	**7**	Duplicate 7
[+]	**14**	Double: 7 + 7 = 14
[1]	**1**	Near-double difference (1)
[+]	**15**	14 + 1 = 15!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Retyping the number to double it (e.g., [7] [ENTER] [7] [+]) instead of recognizing that [ENTER] duplicates Register X into Register Y.

Hands-On Block Remediation: Place matching [7] Digit Blocks in Row 1 and Row 0 of the Place Value Tray, then press [7] [ENTER] and point to the screen: both Y and X already hold 7, so pressing [+] doubles it immediately.

Misconception 2: Confusing '3 groups of 4' with '3 + 4' when transitioning from repeated addition to multiplication.

Hands-On Block Remediation: Model three groups of [4] on the Place Value Tray; have the student key [4] [ENTER] [4] [+] [4] [+] = 12 first, then compare directly with [4] [ENTER] [3] [×] = 12.

Extension Challenge (Student Warm-Up B):

Use the known double 6 + 6 to solve the near-double 6 + 7 (double 6, then add 1).

Expected Answer: 13 (6 + 6 = 12, then 12 + 1 = 13).

Doubles, Near-Doubles & Equal Groups on the Place Value Tray

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: TWIN-ROW DOUBLES & NEAR-DOUBLES STRATEGIES

- Placing matching Digit Blocks in Row 1 (top) and Row 0 (bottom) of the Place Value Tray models a physical Double (n + n).
- Near-Doubles: When two addends differ by 1 (such as 7 + 8), double the smaller number first (7 + 7 = 14) and add 1 more (14 + 1 = 15).
- On the SC-6, pressing [ENTER] duplicates the number on Line X onto Line Y so pressing [+] checks any double in 3 keystrokes.

Key Sequence: `6` `ENTER` `+`

Investigation 1: Two matching rows on the Place Value Tray each hold 8. What is double 8 (8 + 8)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+, −, or ×):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 2: Use the double 8 + 8 to solve the near-double 8 + 9 (double 8, then add 1).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+, −, or ×):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 3: A carton has two equal rows of 9 eggs. What is double 9 (9 + 9)?

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+, −, or ×):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 4: Use the double 5 + 5 to solve the near-double 5 + 6 (double 5, then add 1).

1. Draw Your Blocks on the Tray:

2. Number Bond & Two Quantities:

First Quantity (Top Row):

Second Quantity (Bottom Row):

Combine (+, −, or ×):

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Unit 04: Doubles, Near-Doubles & Equal Groups on the Place Value Tray — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades 1–2

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Place [7] in both rows of your Place Value Tray to build a twin-row double. What is the sum of two 7s (7 + 7)?

Core Solution: 14 ([1] Ten and [4] Ones on the tray; 7 + 7 = 14).

Core Check Keys: 7 ENTER +

Extension Challenge (Warm-Up B): Use the known double 6 + 6 to solve the near-double 6 + 7 (double 6, then add 1).

Extension Solution: 13 (6 + 6 = 12, then 12 + 1 = 13).

Ext. Check Keys: 6 ENTER + 1 +

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Two matching rows on the Place Value Tray each hold 8. What is double 8 (8 + 8)?

Expected Answer: 16

SC-6 Verification Keys: 8 ENTER +

Investigation 2: Use the double 8 + 8 to solve the near-double 8 + 9 (double 8, then add 1).

Expected Answer: 17

SC-6 Verification Keys: 8 ENTER + 1 +

Investigation 3: A carton has two equal rows of 9 eggs. What is double 9 (9 + 9)?

Expected Answer: 18

SC-6 Verification Keys: 9 ENTER +

Investigation 4: Use the double 5 + 5 to solve the near-double 5 + 6 (double 5, then add 1).

Expected Answer: 11

SC-6 Verification Keys: 5 ENTER + 1 +

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.